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Predicting the diabetic foot in patients with type 2 diabetes mellitus based on machine learning

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Abstract

Background: Diabetic foot (DF) is a severe complication of type 2 diabetes mellitus (T2DM), contributing to significant morbidity and healthcare costs globally. Early prediction and intervention are critical for preventing amputations and improving patient outcomes. However, traditional statistical methods lack the capacity to handle high-dimensional clinical data and identify optimal predictive features. This study aimed to develop and validate machine learning models for DF risk prediction using feature selection strategies based on binary logistic regression and information theory.

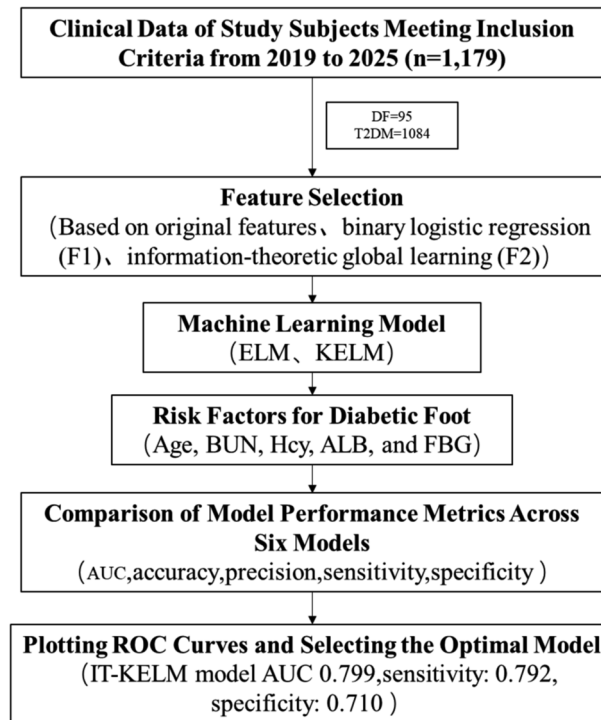
Methods: A retrospective cohort of 1,179 patients (95 DF cases, 1,084 T2DM controls) was analyzed using clinical and biochemical data from 2019 to 2025. Three data sets were constructed: (1) original features; (2) features selected via binary logistic regression (F1); and (3) features selected via information-theoretic global learning (F2). Six models—extreme learning machine (ELM), kernel extreme learning machine (KELM), and their variants trained on the three data sets—were evaluated using fivefold cross-validation. Performance metrics included area under the receiver operating characteristic curve (AUC), sensitivity, specificity, and computational efficiency.

Results: Age, blood-urea-nitrogen (BUN), homocysteine (Hcy), albumin (ALB), and fasting blood glucose (FBG) were identified as independent DF risk factors. The information theory-based KELM (IT-KELM) model achieved the highest AUC of 0.799 (sensitivity: 0.792 and specificity: 0.710) on F2, outperforming other models. Feature selection improved predictive accuracy while reducing computational time, with IT-KELM requiring 0.138 s for training and 0.0023 s for testing. The SHAP summary dot plot and bar chart revealed that the top five features contributing to the model were TP, RBC, ALB, BMI and HB.

Conclusions: Integrating information theory with KELM enhances DF risk prediction by optimizing feature subsets and leveraging nonlinear kernel mapping. The IT-KELM model demonstrates robust diagnostic performance and clinical feasibility for early DF screening. Future multi-center studies are needed to validate generalizability and refine model interpretability in real-world settings. This approach provides a cost-effective tool for precision medicine in diabetes care.

Keywords: Type 2 diabetes mellitus, Diabetic foot, Feature selection, Machine learning, Extreme learning machine, Information theory



Graphical abstract**Introduction**

Type 2 diabetes mellitus (T2DM), as one of the most common chronic diseases, is estimated to have affected 537 million people globally in 2021, with projections indicating this number will reach 643 million by 2030 and 783 million by 2045 [1, 2]. Hyperglycemia associated with diabetes can lead to vascular complications, among which DF stands out as one of the most severe, with a global prevalence rate of about 6.3% [3].

DF can result in significant pain, gangrene, and ultimately, amputation, imposing substantial financial burdens and physical suffering on patients and their families [4]. This study aims to identify high-risk factors for DF and develop a machine learning classification model based on patient biomarker data, offering an economical early warning system that aids clinical decision-making.

Research has indicated that factors such as age, blood-urea-nitrogen (BUN), serum albumin (ALB), fasting blood glucose (FBG), and body mass index (BMI) are risk factors contributing to the development of DF [5, 6]. Current research models on DF predominantly explore the relationship between these risk factors and disease onset, with direct application in predicting the incidence of diabetic foot being relatively scarce. Traditional medical statistical methods are no longer sufficient for handling the vast amounts of patient data in the medical field today. Therefore, it is proposed to utilize computerized approaches to streamline data dimensions. By investigating the correlations among features within data sets, feature redundancy, and their impacts,

an optimal subset of features can be identified. While machine learning algorithms have been applied in exploring diseases, such as cancer and pneumonia [7–9], there is a lack of application of algorithms like extreme learning machine (ELM) in predicting diabetic foot. Hence, this study aims to explore the use of ELM and its improved variants under the support of information theory and binary logistic regression analysis for feature selection in the prediction of diabetic foot incidence.

Results

Previous studies have shown that there is no statistically significant difference in sample sizes between the DM group and the DF group. However, compared with the DM group, the DF group showed significantly higher levels of Age, Scr, BUN, Hcy and FBG ($p < 0.05$). In addition, ALB levels were significantly lower in the DF group than in the DM group ($p = 0.006$), indicating a statistically significant difference as summarized in Table 1.

The binary logistic regression analysis identified Age, BUN, Hcy, ALB, and FBG as significant risk factors for DF. These variables were designated as the logistic regression-derived feature subset (F1) for this study. To further refine feature selection, an information-theoretic global learning method was applied to the original patient data

Table 1 Comparison of general characteristics and biochemical indicators

Factor	DF (n = 95)	DM (n = 1084)	t/X ² /Z	p value
Gender (male/female)	56/39	680/404	0.533	0.465
Age (years)	62.94 ± 11.86	59.97 ± 10.45	−2.61	0.009
Duration of diabetes (years)	7.21 (1.12,25.21)	8.59 (0.23,35.55)	−1.562	0.118
Smoking history (yes/no)	54 (56.8)	669 (61.7)	0.647	0.421
Systolic blood pressure (mmHg)	136.99 ± 20.57	136.52 ± 17.3	0.216	0.829
Diastolic blood pressure (mmHg)	76.9 ± 10.44	75.06 ± 10.6	1.621	0.105
C-peptide (ng/ml)	2.31 ± 1.07	2.17 ± 1.44	−0.925	0.355
Glycated hemoglobin (%)	8.95 ± 2.42	8.6 ± 2.09	−1.529	0.127
Body mass index (kg/m ²)	25.04 ± 2.92	25.32 ± 3.44	0.745	0.457
Fasting blood glucose (mmol/L)	10.04 ± 4.12	8.08 ± 2.6	−4.535	< 0.001
Homocysteine (μmol/L)	14.16 ± 5.23	11.49 ± 4.64	−4.788	< 0.001
White blood cell count (× 10 ⁹ /L)	6.72 ± 2.16	6.55 ± 1.94	−0.807	0.42
Neutrophils (× 10 ⁹ /L)	4.23 ± 1.81	4.01 ± 1.35	−1.172	0.244
Red blood cell count (× 10 ¹² /L)	4.45 ± 0.7	4.52 ± 0.54	0.966	0.366
Hemoglobin (g/L)	132.86 ± 18.96	135.84 ± 16.81	1.474	0.144
Platelet count (× 10 ⁹ /L)	185.41 ± 62.89	189.46 ± 61.47	0.611	0.541
Total cholesterol (mmol/L)	4.39 ± 1.2	4.27 ± 1.15	−0.944	0.345
Triglycerides (mmol/L)	2.12 ± 1.69	2.05 ± 1.61	−0.396	0.692
High-density lipoprotein (mmol/L)	1.09 ± 0.27	1.1 ± 0.31	0.474	0.635
Low-density lipoprotein (mmol/L)	2.82 ± 1.03	2.67 ± 0.96	−1.445	0.149
Alanine aminotransferase (U/L)	23.43 ± 21.54	21.92 ± 14.9	−0.903	0.367
Aspartate aminotransferase (U/L)	22.41 ± 17.01	20.44 ± 10.79	−1.109	0.27
Serum total protein (g/L)	65.15 ± 5.91	65.29 ± 5.82	0.236	0.813
Albumin (g/L)	40.76 ± 4.64	42.02 ± 4.18	2.776	0.006
Serum creatinine (μmol/L)	75.96 ± 31.08	67.54 ± 25.31	−2.556	0.012
Blood-urea-nitrogen (μmol/L)	6.09 ± 2.25	5.47 ± 1.89	−2.996	0.003

Table 2 Model classification performance with original data vs. feature selection

	ELM	KELM	Logistic-ELM	Logistic-KELM	Information theory-ELM	Information theory-KELM
HammingLoss	0.556	0.547	0.546	0.540	0.537	0.532
Example-based accuracy	0.888	0.905	0.892	0.909	0.907	0.916
Label-based accuracy	0.489	0.547	0.526	0.576	0.545	0.582
Label-based recall	0.543	0.599	0.590	0.606	0.613	0.628
Label-based F-measure	0.553	0.634	0.613	0.523	0.640	0.677
MicroF1-Measure	0.872	0.895	0.888	0.899	0.905	0.916
Average_Precision	0.934	0.953	0.944	0.950	0.956	0.958
AUC	0.687	0.756	0.727	0.757	0.726	0.799
Accuracy	0.874	0.905	0.888	0.909	0.893	0.904
TrainingTime (s)	0.0938	0.03	0.125	0.095	0.148	0.138
TestingTime (s)	0	0.004	0.005	0.0082	0	0.0023

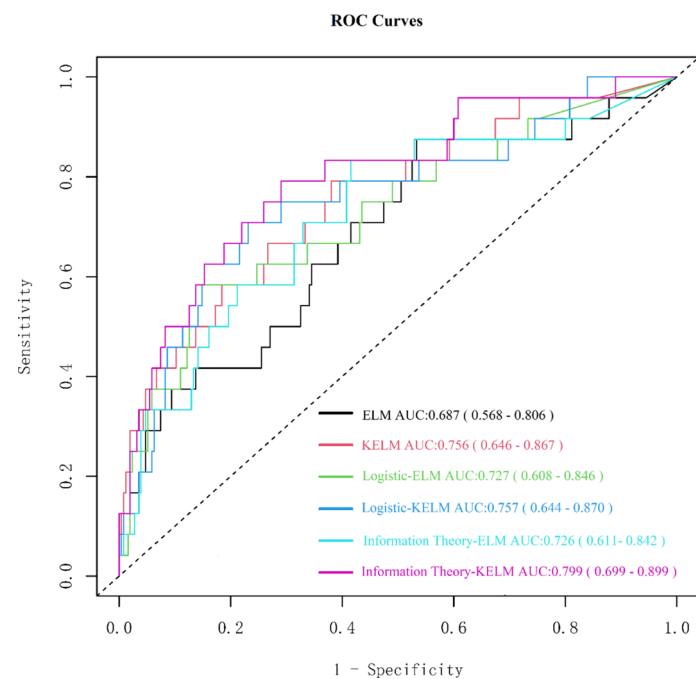


Fig. 1 Comparison of ROC curves and AUC metrics for ELM, KELM, logistic-ELM, logistic-KELM, information theory-ELM, information theory-KELM models

set, resulting in a dimensionality-reduced feature subset (F2). Both subsets (F1 and F2) were subsequently used as input attributes for machine learning models, specifically the ELM and KELM. The corresponding feature values were extracted from the original data set, and models were trained on the training set and validated on the test set. The performance metrics of these models are summarized in Table 2.

Based on the predicted probabilities from six models, we calculated and plotted the ROC curves and AUC metrics. The results indicated that the IT-KELM model achieved the highest AUC value of 0.799, with a sensitivity of 0.792 and specificity of 0.710. This demonstrates that the IT-KELM model possesses significant diagnostic value for predicting DF (Fig. 1).

SHAP (SHapley Additive exPlanations) values were employed to provide a global interpretation of the model's predictions. The SHAP summary dot plot and bar chart revealed that the top five features contributing to the model were TP, RBC, ALB, BMI and HB (Fig. 2).

Discussion

DF is a common chronic complication in patients with T2DM and is strongly associated with higher rates of amputation [10]. Currently, there is a lack of effective pharmacological interventions to improve prognosis in DF patients, making it essential to investigate related risk factors and develop diagnostic prediction models. In this study, we utilized routinely collected clinical and biochemical laboratory data from T2DM patients to explore the risk factors for DF and establish a predictive model. Our findings identified age, BUN, Hcy, ALB, and FBG as independent risk factors for DF.

In this study, age distribution showed a significant association with DF. The mean age differed notably between the two groups (62.94 years in the DF group vs. 59.97 years in the T2DM group), indicating a significant difference between the groups. This finding is consistent with numerous studies in which advanced age has been identified as an important risk factor for DF [11]. Paras M et al. reported that individuals over 55 years of age have the highest predisposition to DF, followed by those aged 36–55 and 26–35 years [12]. Similarly, Banik PC et al. found that being under 50 years of age may confer a protective effect against DF, whereas being over 50 is positively correlated with its occurrence [13].

In vitro studies have shown that elevated Hcy levels can cause endothelial cell damage either directly through cytotoxic effects or indirectly via oxidative stress, leading to microvascular occlusive arteriosclerosis and subsequent development of DF [14, 15]. Abraham et al. reported that elevated Hcy levels are predictive of DF onset, and

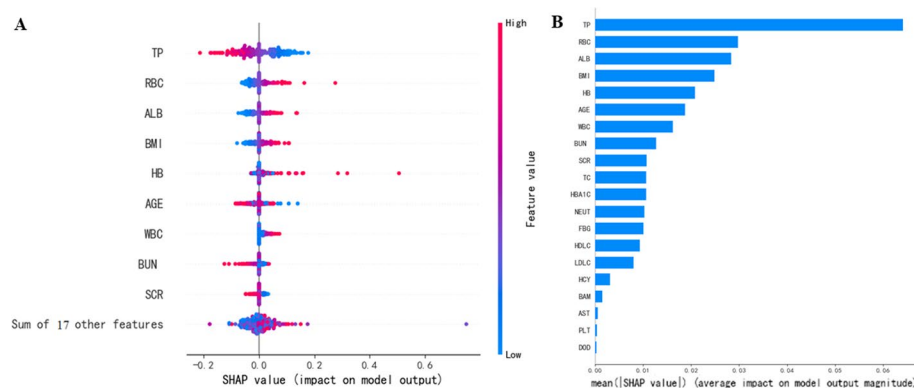


Fig. 2 SHAP-based feature importance analysis for predicting DF. **A** SHAP summary dot plot. The SHAP summary dot plot shows the contribution of each feature. Each point represents a patient's feature, with the color indicating the size of the feature value, red representing higher values and blue representing lower values. The vertical Stacking of points reflects the sample density. **B** SHAP summary bar chart. The SHAP summary bar chart displays the average contribution of each feature to the model's output, sorted by the magnitude of contribution

although vitamin supplementation can reduce Hcy levels, it does not significantly lower the incidence of cardiovascular events [16]. Moreover, high Hcy levels have been linked to the occurrence of vascular-type DF [17].

Our results also demonstrated a significant association between admission FBG levels and the development of DF. Hyperglycemia has long been recognized as a risk factor for DF due to its role in the progression of microvascular complications and peripheral neuropathy [18]. This finding aligns with other reports suggesting that poor glycemic control is a major contributor to DF-related complications [19–21]. This study further confirmed a significant independent association between FBG levels and the risk of DF. Consistent with our findings, Moodley RM et al. reported no significant difference in HbA1c levels between DF and non-DF patients [22].

This study established and validated a DF prediction model for predicting the risk of DF in diabetic patients. The model integrates patient baseline characteristics and clinical blood test indicators, employing binary logistic regression, information theory, ELM, and KELM algorithms to construct predictive frameworks. These models aim to provide clinical decision support for early diagnosis and intervention in DF. As an artificial intelligence technique, machine learning offers novel insights into precision prediction by leveraging feature learning from input data. This encompasses traditional methods, such as logistic regression and decision trees, as well as ensemble methods, such as random forest, light GBM, and XGBoost. Prior studies have demonstrated the efficacy of these models in disease prediction [23, 24]. For instance, Du et al. [25] utilized back-propagation neural networks, logistic regression, and random forest algorithms with four risk factors—including age and free prostate-specific antigen—to establish a prostate cancer diagnostic model. Their results highlighted that machine learning-based multi-factor predictive models outperform single-factor models, aiding clinicians in diagnostic and therapeutic decision-making while reducing unnecessary examinations.

In this study, we applied three modeling approaches—direct prediction, binary logistic regression-based prediction, and information theory-based feature selection prediction—to the same data set. As shown in Table 2, KELM (AUC:0.756) outperformed ELM (AUC:0.687), because KELM replaces the random mapping process of ELM with a kernel function, enabling superior classification in the high-dimensional separable space mapped by the kernel function.

Binary logistic regression analysis identified Age, BUN, Hcy, ALB, and FBG as significant risk factors for DF, consistent with prior literature. These five features were used as a filtered subset for prediction in this study. Experimental results demonstrated that logistic–ELM achieved superior performance compared to traditional ELM using all features, with an AUC of 0.727, a Micro-F1 score of 0.888, and an Average Precision of 0.944. Feature selection enhanced prediction accuracy by identifying optimal subsets. In addition, information theory-based feature selection identified the top five features as TP, RBC, ALB, HB, and BMI, which differed slightly from logistic regression results. From a computational correlation perspective, DF occurrence was more closely associated with this subset. However, while models using information theory-based ELM and KELM outperformed original algorithms and logistic regression, their interpretability was limited due to discrepancies with established clinical risk factors. This discrepancy stems from the fundamental differences between the two methodologies. Logistic

regression is a statistical approach based on linear assumptions, designed to identify variables that exhibit independent statistical associations with the outcome after adjusting for the effects of other covariates. In contrast, information-theoretic methods are non-parametric and focus on quantifying the overall information shared between features and the outcome while explicitly accounting for redundancy among features. Their objective is to identify an optimal feature subset that maximizes predictive performance, rather than solely prioritizing individual statistical significance.

Although the specific selected features differ, both approaches underscore the central role of nutritional status—ALB was selected in both models. The feature set identified by the information-theoretic method—TP, RBC, HB, ALB and BMI—collectively reflects the patient's systemic nutritional status, anemia severity, and tissue oxygenation capacity. This suggests that, beyond conventional metabolic and vascular risk factors, the patient's overall metabolic and hematological health may represent a critical dimension in predicting DF risk.

For instance, anemia (indicated by low RBC and HB) and hypoalbuminemia (low TP and ALB) can impair tissue perfusion and compromise wound healing, while BMI may serve as a proxy for underlying metabolic health. The information-theoretic approach appears better equipped to capture the synergistic predictive information embedded within this combination of biomarkers. Rather than indicating contradiction, this divergence highlights the advantage of information-theoretic methods in handling high-dimensional, nonlinear clinical data, enabling the discovery of clinically meaningful feature interactions that might be overlooked by traditional statistical models. These findings also suggest that, in addition to routinely monitored parameters, such as glycemic control and renal function, clinicians should comprehensively evaluate patients' nutritional and hematological profiles when assessing DF risk.

This study did not aim to refute logistic regression-derived risk factors but to explore methods to enhance predictive accuracy. Both logistic regression- and information theory-based feature selection achieved comparable results (e.g., logistic-KELM: AUC:0.757; information theory-KELM: AUC:0.799), both surpassing the baseline ELM (AUC:0.687) and KELM (AUC:0.756) models. These approaches effectively identified DF patients, providing clinicians with early diagnostic and intervention guidance. In ELM frameworks, single-hidden-layer feedforward neural networks (SLFNNs) achieve faster computation than traditional machine learning algorithms while maintaining equivalent learning accuracy. By randomly initializing input weights and hidden-layer biases, ELM directly solves linear equations for output matrices, streamlining training. KELM further improves upon ELM by replacing random mapping with kernel mapping, eliminating the need to explicitly define mapping functions or optimize hidden neuron counts. This enhances generalization and stability, addressing limitations of random weight initialization in ELM. Given the study's focus on developing a simple DF screening tool for clinical use, both traditional ELM/KELM and feature-selected ELM models are viable. All six models achieved real-time prediction (< 1 s), enabling immediate clinical application in primary healthcare settings and health examination systems.

The study has three key strengths: (1) the ELM models using five selected features are clinically accessible, validated for high relevance to DF, and demonstrate robust classification performance; (2) this work bridges the gap in DF research by applying machine

learning-based prediction models, offering methodological insights for future studies; and (3) the proposed direct and feature-selected prediction approaches enable cost-effective, early DF screening, supporting timely clinical interventions.

A key limitation of the present study is the lack of robust temporal validation and thorough model calibration. The recent work by Ding et al. [26] provides a valuable roadmap for future advancements: they successfully leveraged 5-year electronic health record (EHR) cohort data with a large language multimodal model to predict incident type 2 diabetes, underscoring the importance of rigorous temporal validation frameworks, advanced feature fusion strategies, and meticulous model calibration. Their findings highlight the critical role of modeling temporal dynamics and ensuring model robustness—insights that strongly support the incorporation of longitudinal data and more sophisticated time-series modeling approaches in future DF prediction research. In addition, our models have not undergone external validation. To enhance generalizability and predictive accuracy while mitigating the influence of site-specific biases and confounding factors, future studies should validate these models in multi-center, large-scale cohorts and conduct exploratory analyses to establish clinically meaningful risk stratification schemes and decision thresholds.

Conclusion

This study developed and validated a DF prediction model using ELM algorithms, incorporating original patient data and features selected via logistic regression and information theory. The model achieves high diagnostic accuracy (AUC:0.799), offering a practical tool for early DF risk assessment in clinical decision-making, community healthcare, and home care settings. While the methodology demonstrates strong stability and efficiency, further validation with expanded data sets and multi-center studies is required to confirm its broader clinical utility and applicability. This research underscores the potential of machine learning to enhance early DF screening and management, particularly in resource-limited settings.

Materials and methods

Data sources

This study covers the time period from 2019 to 2025, with data sourced from the medical records of diabetic inpatients at Hefei Hospital Affiliated to Anhui Medical University. The inclusion criteria were: patients whose diagnostic information conformed to the "Clinical guidelines for prevention and treatment of T2DM in the elderly in China (2022 Edition)" [27], who had diabetic foot Wagner grade I or higher [28], and whose cases were approved by the hospital's ethics committee. Exclusion criteria included patients with heart, lung, liver, or kidney dysfunction, diabetic ketoacidosis, incomplete medical records, or those with peripheral neuropathy or systemic infections related to DF. The total sample comprised 1,231 patients. After excluding 52 individuals with missing and duplicate data for key variables, the final analytic cohort included 1,179 patients, including 1,084 T2DM patients and 95 DF group patients. A table detailing the patterns of missing data and the handling of outliers for the 52 excluded patients was prepared and provided as Supplementary Material S1.

General information

As this research involves retrospective data, it did not require patient informed consent. All participants involved in the study were clinical researchers from our hospital. Patient data covered three dimensions: socio-demographic characteristics, medical history, and clinical data.

Socio-demographic characteristics: gender, age, and BMI. Medical history: duration of illness, previous history of DF, smoking (at least one cigarette per day for more than half a year), and blood pressure (systolic/diastolic). Clinical data: fasting blood glucose (FBG), glycated hemoglobin, C-peptide, homocysteine (Hcy), white blood cell count, neutrophil count, red blood cell count (RBC), hemoglobin (HB), platelet count, total cholesterol, triglycerides, high-density lipoprotein, low-density lipoprotein, alanine aminotransferase, aspartate aminotransferase, total serum protein (TP), albumin (ALB), serum creatinine (Scr), and blood-urea-nitrogen (BUN).

Methods

The original data set was first analyzed using univariate logistic regression to identify risk factors for DF. Subsequently, an information-theoretic global learning approach was applied to perform feature selection and derive the optimal subset of predictive features. To comprehensively evaluate the predictive performance of ELM and kernel extreme learning machine (KELM) models on the DF data set, three distinct data sets were constructed: (1) the original data set; (2) a data set comprising only the risk factors identified through univariate logistic regression; and (3) a data set consisting of the optimal feature subset selected via the information-theoretic global learning method. The ELM and KELM models, along with four other machine learning algorithms (a total of six models), were trained and evaluated across these three data sets to compare their predictive performance. To more robustly assess the generalization ability of the models, this study employed fivefold cross-validation to mitigate the impact of limited sample size and class imbalance, thereby improving the predictive performance of six machine learning models for diabetic foot prediction.

Previous studies using binary logistic regression analysis have selected the relevant features for predicting DF in patients with T2DM based on the presence or absence of DF as the dependent variable in the data set.

The information theory-based global learning method takes into account both the feature correlation and feature redundancy in DF data [29]. It identifies the optimal feature subset by maximizing the correlation between candidate features and the outcome, thereby achieving dimensionality reduction.

ELM (Huang et al.) [30] theories show that the hidden-layer parameters of single-hidden-layer feedforward networks (SLFNs) needs no tuning and can be generated randomly, independent of the training data set, and the output weights are computed in a single step by employing Generalized Moore–Penrose pseudo-inverse. Due to the random generation of the hidden-layer weights and the biases in ELM, the training speed is much faster compared to the traditional back-propagation (BP) algorithm. ELM has been widely applied in fields, such as image recognition, text classification, and multi-label classification [31, 32].

Given N examples $[x_1, x_2, \dots, x_N]$ and their corresponding targets $[t_1, t_2, \dots, t_N]$. Here, the input vector $X_i = [x_{i1}, x_{i2}, \dots, x_{in}]^T \in R^n$ and its desired output $t_i = [t_{i1}, t_{i2}, \dots, t_{im}] \in R^m$. Here, T stands for transpose. Here, N is the feature dimensional input space, L and m are the number of neurons in the hidden layer and the output layer, respectively. Usually, we assume the number of samples $N > L$. The output function of ELM for generalized SLFNs is

$$f_L(x) = \sum_{i=1}^L \beta_i h_i(x) = h(x) \beta \quad (1)$$

where β_i is the output weight vector between the hidden layer of L nodes to the $N > L$ output nodes and $h(x)$ is ELM nonlinear feature mapping, $h_i(x)$ is the output of the i th hidden node output.

In the second stage of ELM learning, the weights connecting the hidden layer and the output layer, denoted by β , are solved by minimizing the approximation error in the squared error sense:

$$\min_{\beta \in R^{L \times N}} \|H\beta - T\| \quad (2)$$

Different from traditional gradient descent methods, ELM only requires randomly initializing the input weights and hidden-layer biases to obtain a unique hidden-layer output matrix. This transforms the model training into a linear equation $H\beta = T$ solving problem. According to the least squares method, we have:

$$\hat{\beta} = H^+ T \quad (3)$$

where H^+ denotes the Moore–Penrose generalized inverse of matrix H .

KELM [33, 34] replaces the original random mapping process of ELM by introducing a kernel function. Through the implicit mapping of the kernel function, it cleverly computes the inner product between features, mapping low-dimensional non-linearly separable samples to a higher dimensional separable space. Since there is no need to explicitly define the mapping function, it avoids the need to predefine the number of hidden-layer neurons, effectively addressing issues such as generalization and instability of results caused by randomly assigned neuron numbers. Based on the prediction output model of ELM and the KKT conditions, the output of the ELM model is

$$\beta_i = H_i^T \left(\frac{I}{C} + H_i H_i^T \right)^{-1} Y_i \quad (4)$$

By explicating the kernel matrix $HH^T(i, j) = K(x_i, x_j)$, $(\Omega_{ELM})_{ij} = h(x_i) \cdot h(x_j) = K(x_i, x_j)$, the ELM model can be rewritten as

$$f_i(x) = H_i \beta_i = H_i H_i^T \left(\frac{I}{C} + H_i H_i^T \right)^{-1} Y_i = \begin{bmatrix} K(x, x_{i1}) \\ \vdots \\ K(x, x_{im}) \end{bmatrix}^T \left(\frac{I}{C} + \Omega_{iELM} \right)^{-1} Y_i \quad (5)$$

where C is a regularization parameter, I is an identity matrix of dimension N .

$$K(x, x_i) = \exp \left(-\frac{\|x - x_i\|^2}{2\sigma^2} \right) \quad (6)$$

Model establishment and evaluation

The significant risk factors identified through Logistic Regression and feature subsets obtained via an information-theoretic global learning approach were utilized to construct ELM and KELM models for disease prediction. The performance evaluation metrics included ROC curves and AUC metrics, Hamming loss, example-based accuracy, label-based accuracy, label-based recall, label-based f-measure, micro-F1 measure, average precision, and accuracy. In the experiments, the radial basis function (RBF) kernel was selected as the kernel function, and the activation function of the ELM was set to 'radbas'. The number of hidden-layer neurons in the ELM was fixed at 100, the regularization parameter C was set to 50, and the kernel parameter (Kernel_para) was assigned a value of 1.

Statistical analysis

Statistical analyses were performed using SPSS software (version 23.0; IBM Corporation, USA). Normally distributed continuous variables are presented as mean $\pm$ standard deviation ($\bar{X} \pm SD$), and group comparisons were conducted using the independent samples t test. Non-normally distributed continuous variables are expressed as median and interquartile range [M (QL, QU)], and comparisons between groups were performed using the Mann–Whitney U test. Categorical variables are presented as frequency and percentage [n (%)], and the chi-square (χ^2) test was used for intergroup comparisons. Multivariable logistic regression analysis was employed to identify independent risk factors for the development of diabetic foot. All statistical tests were two-sided, with a significance level set at $\alpha=0.05$. A $p < 0.05$ was considered statistically significant. All modeling and analytical procedures were implemented within the MATLAB R2015b software (MathWorks Corporation, USA). The model employed in this study—particularly the KELM—is sensitive to the scale of input features. To mitigate the influence of differing scales across features and ensure training stability and convergence, all continuous variables (e.g., age, BUN, Hcy, ALB, and FBG) were standardized using Z-score normalization prior to model training, defined as $(x - \mu)/\sigma$, where μ is the mean and σ is the standard deviation. Categorical variables (e.g., sex and smoking history) were encoded using one-hot encoding.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s12938-025-01494-2>.

Supplementary material 1.

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None.

Author contributions

HX Z: methodology and writing—original draft; WJ F: data curation, formal analysis, and conceptualization; PP L: supervision; XZ C and SW Y: writing—review and editing. All authors read and approved the final manuscript.

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Data availability

Not applicable.

Declarations**Ethics approval and consent to participate**

Not applicable.

Consent for publication

Not applicable.

Competing interest

The authors declare no competing interest.

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